

SURFACE AREA

Surface area is the sum of the areas of the faces of a three-dimensional figure. You already know how to find the area of a complex figure. You can use this knowledge to find surface area.

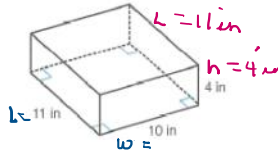
Box has 4 sides, a top, a Bottom

FINDING SURFACE AREA BY ADDING AREAS

Example Find the surface area of the box.

The box, a rectangular prism, has 6 faces. Each face is a rectangle that has its own length and width.

Imagine unfolding the box so that the cardboard lies flat. Make sure you understand how the second drawing can be folded to make the box in the first drawing.



Step 1

Find the area of each rectangle.

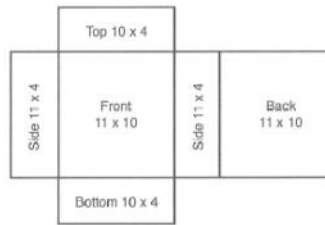
Top:	$10 \times 4 = 40$
Bottom:	$10 \times 4 = 40$
Left side:	$11 \times 4 = 44$
Right side:	$11 \times 4 = 44$
Front:	$11 \times 10 = 110$
Back:	$11 \times 10 = 110$

Step 2

Add the areas of the faces.

40
40
44
44
110
110
$+ 110$
388

The surface area of the box is **388 square inches**.



SA = area of the top & Bottom + area of all the sides

$$SA = 2(11 \times 10) \text{ Top \& Bottom} + 4(11 \times 4)$$

one pair of sides 11×4
one pair of sides 10×4

$$SA = 2(11 \times 10) + 2(11 \times 4) + 2(10 \times 4)$$

$$= 220 + 88 + 80$$

$$SA = 388 \text{ in}^2$$

Surface Area Formulas

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You will be expected to find the surface areas of rectangular prisms, pyramids, cylinders, cones, and spheres. To make your work easier, the GED math formula sheet includes formulas for finding the surface areas of these figures.

Surface Area (SA) Formulas

(p = perimeter of base, B = area of the base,
 h = height, r = radius, s = slant height)

Rectangular/Right Prism	$SA = ph + 2B$	}	think logically
Cylinder	$SA = 2\pi rh + 2\pi r^2$		
Square Pyramid	$SA = \frac{1}{2}ps + B$		
Cone	$SA = \pi r s + \pi r^2$	}	the length of side
Sphere	$SA = 4\pi r^2$		

} know how to use the formula

Remember, a formula uses algebra to write a procedure for doing something. These formulas add the areas of the faces of a figure. You don't have to use formulas, but they may save you time.

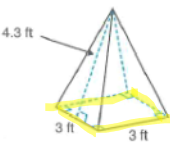
USING A FORMULA TO FIND SURFACE AREA

Example Find the surface area of the pyramid.

Use the formula: $SA = \frac{1}{2}ps + B$

Step 1
Find the facts you need to use the formula.

The slant height s is 4.3 feet.
 Find p , the perimeter of the base:
 $3 + 3 + 3 + 3 = 12$
 Find B , the area of the base: $3^2 = 9$



Step 2
Evaluate the formula.

$$SA = \frac{1}{2}ps + B$$

$$= \frac{1}{2} \cdot 12 \cdot 4.3 + 9$$

$$= 34.8 \text{ sq ft}$$

The surface area is **34.8 sq ft**.

Find surface area
 = total area of the outside
 = area of the base
 + area of the triangles

$$SA = s^2 + 4 \left(\frac{1}{2}bh \right)$$

$$s = 3, b = 3, h = 4.3$$

$$SA = 3^2 + 4 \left[\frac{1}{2}(3)(4.3) \right]$$

$$= 9 + 6(4.3)$$

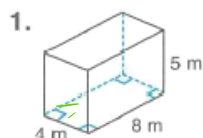
$$= 9 + 25.8$$

$$SA = 34.8 \text{ ft}^2$$

Examples

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Find the surface area of each figure. If necessary, round your answer to the nearest tenth square unit.

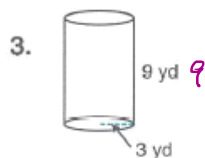


2 tops: $A = 2(4 \times 8)$
~~top & bottom~~ $= 64$
 2 sides: $A = 2(5 \times 8)$
 $= 80$
 2 sides: $A = 2(4 \times 5)$
 $= 40$
 $SA = 64 + 80 + 40$
 $SA = 184 \text{ m}^2$

$$C = \pi d$$

$$C = 2\pi r$$

Because
 $d = 2r$



top & Bottom:
 2 (area of a circle)
 $A = 2\pi r^2$
 $= 2\pi(3^2)$
 $= 18\pi$
 $A = 56.52$

Side of the cylinder:

$$w = C = \pi d \quad \begin{matrix} h = 9 \\ d = 6 \end{matrix}$$

$$A = hC$$

$$= 9(\pi 6)$$

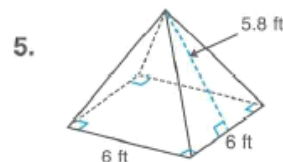
$$= 54\pi$$

$$A = 169.56$$

$$SA = 56.52$$

$$+ 169.56$$

$$SA = 226.1 \text{ yd}^2$$



square pyramid
 4 triangular sides

$$SA = \text{Base} + 4\Delta s$$

$$= 6^2 + 4\left(\frac{1}{2}bh\right)$$

$$= 6^2 + 4\left[\frac{1}{2}(6)(5.8)\right]$$

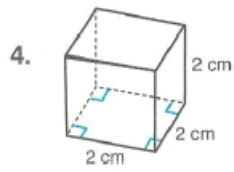
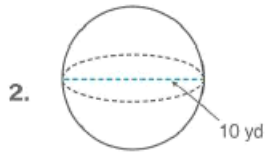
$$= 36 + 12(5.8)$$

$$SA = 105.6 \text{ ft}^2$$

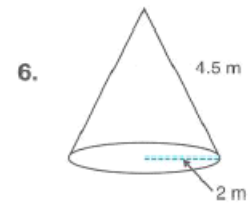
Examples, cont'd...

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Sphere $SA = 4\pi r^2$



Cone $SA = \pi r s + \pi r^2$

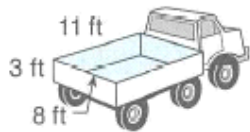


And more Examples

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Solve as directed.

7. The illustration shows a truck bed with no top. What is the surface area of the inside of the truck bed?



8. Crunchy Oats cereal comes in a box that measures 8 by 11 by 2 inches. Bran Crisps cereal comes in a box that measures 7 by 10 by 3 inches. (*Hint:* Make a drawing with labels.)
- Which box has a larger surface area?
 - Which box has a greater volume?

CHOOSING AREA, PERIMETER, VOLUME, OR SURFACE AREA

You know how to find the perimeter, area, volume, and surface area of various figures. But knowing which to solve for takes practice.

Remember: **Perimeter** is the distance around a flat figure. **Area** is the space inside a flat figure. **Volume** is the space inside a three-dimensional figure. **Surface area** is the sum of the areas of the faces of a three-dimensional figure.

For each problem, select whether to solve for perimeter, area, volume, or surface area. Then solve the problem. Round to the nearest whole unit if necessary.

1. How many cubic feet does the moving truck shown in the ad hold?

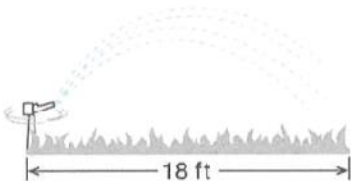


Solve for:

perimeter
area
volume
surface area

Solution: _____

2. A revolving sprinkler sprays a lawn for a distance of 18 feet. How much ground is covered by the sprinkler as it makes 1 complete revolution?



Solve for:

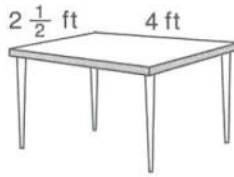
perimeter
area
volume
surface area

Solution: _____

More Examples

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3. How many feet of chrome edging are needed to finish a rectangular kitchen table?



Solve for:

perimeter
area
volume
surface area

Solution: _____

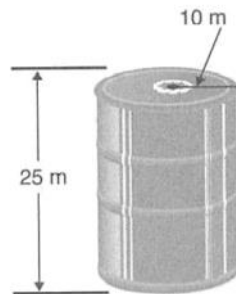
6. A public school wants to install rubber padding over a rectangular playground space that measures 22 feet by 15 feet. The padding comes in square tiles that measure 1 foot by 1 foot. How many tiles are needed?

Solve for:

perimeter
area
volume
surface area

Solution: _____

4. To prevent leakage, an oil storage tank is coated with a sealant. How many square meters will need to be coated to cover the container?

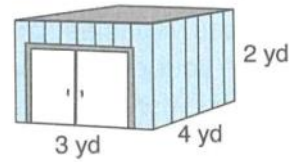


Solve for:

perimeter
area
volume
surface area

Solution: _____

7. A storage unit has the dimensions shown in the drawing. If the top and four sides of the unit are repainted, how many square yards will be painted?



Solve for:

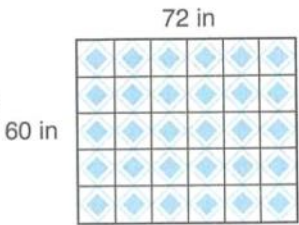
perimeter
area
volume
surface area

Solution: _____

More Examples, cont'd...

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5. Marisol wants to put a satin border on a quilt she made. The drawing shows the dimensions of the quilt. How many inches long is the border?

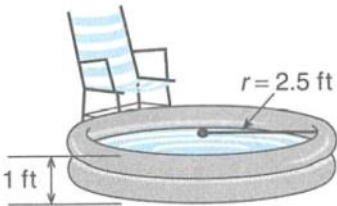


Solve for:

perimeter
area
volume
surface area

Solution: _____

8. How many cubic feet of water will the circular wading pool hold?



Solve for:

perimeter
area
volume
surface area

Solution: _____